

Renewable electricity is associated with wholesale price resilience in sustainable Iberian power markets

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Abstract

Sustainable electricity transitions are often assessed through emissions and generation shares. Their social and economic viability also depends on how power markets transmit fossil-fuel price shocks. This paper studies the Iberian day-ahead market, where Spain and Portugal combine high renewable penetration, internal market coupling and exposure to the 2021–2023 gas-price shock. Using hourly OMIE prices and Spain-Portugal capacity files, REData and REN generation and demand records, and MIBGAS day-ahead gas prices, we build matched daily samples for 2019–2025. The analysis connects three outcomes that are usually examined separately: wholesale price levels, gas-price pass-through and internal MIBEL price dispersion. Renewable generation is negatively associated with day-ahead prices. The total estimate is -0.987 EUR/MWh per percentage point, while the estimate conditional on gas generation is -0.699 EUR/MWh. Renewable penetration is also associated with weaker within-month gas-price pass-through, with a renewable-gas interaction of -0.438 . A four-period check shows a negative interaction before the Iberian Exception, a positive interaction during the formal policy window and a return to a negative interaction after 2023. These estimates describe temporal heterogeneity, not a causal policy effect. Internal Spain-Portugal commercial capacity is associated with narrower hourly absolute price spreads, at -2.924 EUR/MWh per additional GW. The estimates identify conditional associations, not causal effects. They document a price-based pattern under specific market and policy conditions.

Keywords Renewable electricity · Price resilience · Gas-price pass-through · Sustainable energy systems · Market coupling · MIBEL

1 Introduction

Europe's electricity transition is no longer a question of generation mix alone. It is also a question of whether low-carbon power systems can remain affordable, politically durable and socially acceptable when fossil-fuel markets are unstable. During the 2021–2023 energy crisis, wholesale power markets became a visible transmission channel between imported gas prices, public compensation schemes, industrial costs and household bills. A sustainable power system can therefore be judged not only by its emissions profile, but also by the way its market

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design transmits or cushions fuel-price shocks. This paper studies that wholesale-price channel. It does not estimate aggregate economic resilience; instead, it asks whether renewable penetration and internal market coupling are associated with lower exposure of day-ahead prices to gas-market stress.

Wholesale price resilience is necessary but not sufficient for wider socio-economic resilience. Retail exposure also depends on contracts, regulated tariffs, taxes, hedging and public compensation. Lower wholesale prices may therefore not lead to proportionate reductions in consumer bills or energy poverty. The distribution of benefits can also differ across consumer groups. This study examines one market-level component of resilience. It does not estimate retail affordability, distributional outcomes or energy justice.

We use wholesale price resilience in a narrow sense. It refers to lower exposure of day-ahead prices to changes in gas prices. This is not the same as power-system resilience. That concept usually concerns continuity of supply and recovery after disruption [2, 38]. We also examine price differences between Spain and Portugal. Average price levels provide supporting evidence. These measures do not capture physical reliability or retail affordability.

The Iberian electricity market is a useful setting for this question. Spain and Portugal operate within MIBEL, a common day-ahead market with two national bidding zones and explicit internal interconnection constraints. Both systems have large renewable shares, but their generation mixes differ: Spain has a larger solar base and a larger power system, while Portugal relies more heavily on hydro. The peninsula is also weakly connected to the rest of continental Europe. That external constraint motivates the case, but the empirical test here is narrower and cleaner: the paper studies internal Spain-Portugal price integration within MIBEL, not Iberia-France integration.

The existing merit-order literature has shown that wind and solar generation often lower spot prices [7–9, 15, 20, 23, 32, 42, 44]. That evidence is important but incomplete for a crisis-prone sustainable energy system. Lower average prices do not necessarily imply lower fuel-price exposure, smaller price dispersion between coupled zones, or reduced volatility under scarcity conditions. Gas-fired units can still set marginal prices, policy interventions can change bidding incentives, and transmission constraints can separate neighbouring zones even when renewable output is high. The central empirical issue is therefore not simply whether renewables lower prices, but whether they alter the gas-price sensitivity of prices, and whether internal market coupling narrows price gaps when the system is stressed.

Previous studies usually examine these outcomes separately. Merit-order studies focus on average electricity prices. Fuel-market studies examine price transmission. Market-integration studies focus on price convergence across regions. This separation makes it difficult to see whether lower average prices also coincide with weaker gas-price exposure or smaller internal price gaps.

This paper examines the three outcomes in the same market and period. It first estimates the association between renewable generation and day-ahead prices. It then tests whether gas-price pass-through varies with renewable penetration. Finally, it examines the association between internal commercial capacity and Spain–Portugal price differences. The joint analysis does not identify a common causal mechanism. It provides a consistent comparison under the same market and policy conditions.

2 Institutional background

MIBEL gives the paper a paired-market design. The two countries share the same organised Iberian day-ahead market, while remaining separate bidding zones linked by commercial interconnection capacity. When the available capacity between Spain and Portugal is sufficient, hourly prices converge. When it is scarce, the market can split and the two zones clear at different prices. The daily mean of hourly absolute Spain-Portugal price differences is therefore used as the main indicator of internal price dispersion. This differs from broader studies of European market integration, where price convergence is shaped by several borders, heterogeneous regulatory regimes and different market-coupling arrangements [4, 6, 17, 45].

The two systems differ in scale and generation mix. In the raw daily data, Spain generated about 259 TWh in 2019, compared with 49 TWh in Portugal. The renewable share rose from 37.7% to 55.5% in Spain between 2019

and 2025. In Portugal, it rose from 53.0% to 74.3%. Portugal also had a higher hydro share in both years. Wind and solar shares approached 40% in both countries by 2025. Their annual average day-ahead prices remained close because the markets are coupled: 47.68 and 47.87 EUR/MWh in 2019, and 65.16 and 66.06 EUR/MWh in 2025, for Spain and Portugal respectively. These similarities in price conditions, together with differences in generation mix, motivate the paired-country analysis and the country-specific checks.

The Iberian Peninsula's weak external connection with France and continental Europe is relevant for interpretation, but it is not the empirical object of this paper. EU interconnection policy treats cross-border capacity as a condition for competition, security of supply and renewable integration [12]. This background helps explain why the Iberian case is policy-relevant. The regressions below, however, test the internal Spain-Portugal channel only: whether available MIBEL capacity is associated with smaller hourly price gaps inside the Iberian market.

The gas crisis added a second institutional layer. Under marginal pricing, high gas prices can affect electricity prices even when the average renewable share is large. Combined-cycle plants may still set the market-clearing price. Spain and Portugal responded with the Iberian Exception. The temporary mechanism applied a regulated gas-cost reference to eligible thermal generation. Generators received compensation for the gap between that reference and the market gas price. The mechanism changed how gas costs entered bids and how those costs were recovered. It also affected cross-border incentives and compensation flows [19, 26]. The observed market-clearing price therefore did not capture every cost borne by consumers during the intervention [13]. The paper treats the formal mechanism period as a policy-intervention window, not as an ordinary market episode.

3 Literature and hypotheses

The review covers renewable generation and price levels, gas-price pass-through and internal MIBEL price integration.

3.1 Renewables and wholesale electricity prices

The first mechanism is the merit-order channel. Wind and solar have very low short-run marginal costs. An increase in their output shifts residual demand away from higher-cost generators and may lower the market-clearing price. Sensfuss et al. [42] documented this mechanism in Germany. Gelabert et al. citegelabert2011spanish provided ex-post evidence for renewable generation and cogeneration in Spain. Later German studies estimated the contributions of wind and solar together [8] and of wind generation separately [23]. Evidence from the Netherlands and Italy shows that the size of the price association varies across markets and technologies [7, 32]. Woo et al. [44] examined a separate outcome: the association of wind generation with both the level and variance of spot prices in Texas.

The size of the merit-order association depends on the generation displaced at the margin. The price reduction can be larger when additional renewable output replaces gas- or coal-fired generation and smaller when lower-cost units are already marginal. Technology and timing also matter. Solar output is concentrated in daylight hours, wind output is weather-dependent, and hydro dispatch reflects reservoir conditions and the opportunity cost of stored water. At higher renewable penetration, correlated wind or solar production can also reduce the market value of additional output [20]. Storage, demand response, interconnection and flexible generation may shift renewable electricity across hours or locations, but their moderating roles depend on system conditions [16, 25, 27]. These mechanisms motivate the expected negative association in H1, but they are not separately identified in the present models.

This motivates the first hypothesis:

H1. A higher renewable generation share is associated with lower day-ahead electricity prices in Spain and Portugal.

3.2 Gas-price pass-through and price resilience

The second mechanism concerns gas-to-power pass-through. In a marginal-pricing market, the accepted offer of the most expensive generator needed to meet demand sets the market-clearing price. When a gas-fired plant is marginal, a change in its fuel cost can therefore affect the price received by all cleared generators. Mohammadi [30] separated long-run relations between electricity prices and fuel costs from their short-run dynamics. A related study tested integration between electricity markets and their main fuel-source markets [31]. More recent European evidence asks a different question. It shows that gas-fired generation can set electricity prices much more often than its generation share alone would suggest [46]. A power system may therefore remain exposed to gas-price movements even when gas supplies a minority of total electricity.

Natural gas is the focal fuel for four reasons. Gas-fired generation can be marginal in both Iberian bidding zones. The sample also covers the 2021–2023 gas-price shock. The Iberian Exception directly changed the gas-cost reference used by eligible thermal generators. Finally, the MIBGAS PVB day-ahead price provides a common daily fuel-price benchmark for Spain and Portugal. Coal and oil prices do not correspond as closely to this market mechanism or policy episode. The analysis therefore focuses on the fuel-price channel most directly relevant to Iberian marginal pricing and the policy episode studied here.

The strength of this pass-through is conditional. It depends on how often gas-fired plants are marginal, their thermal efficiency and carbon costs, and the availability of lower-cost generation. Higher wind and solar output may reduce gas-price exposure when it displaces gas-fired generation from the margin. This moderation is not automatic. Low renewable or hydro availability, high demand, congestion and scarcity may leave gas-fired units at the margin. Policy intervention can also change how fuel costs enter bids and how they are recovered. The wider macroeconomic literature explains why energy-price shocks matter for spending, inflation and output [18, 24], but the present analysis tests the narrower wholesale electricity-market channel.

H2a: *Higher natural-gas prices are associated with higher day-ahead electricity prices.*

H2b: *Higher natural-gas prices are associated with greater within-day price volatility.*

H3a: *Higher renewable penetration is associated with weaker pass-through of natural-gas prices into day-ahead electricity prices.*

H3b: *The association between wind and solar generation and within-day price volatility varies with natural-gas price conditions.*

3.3 Internal MIBEL price integration

The third mechanism is spatial. If one bidding zone faces a local imbalance, available cross-border capacity allows the neighbouring zone to arbitrage the difference. Price separation should therefore shrink when internal commercial capacity is larger, provided the capacity is not absorbed by other constraints. This channel is especially relevant in high-renewable systems, where output is weather-driven and geographically uneven. It also connects the paper to market-design work on price signals, congestion management, adequacy and flexibility [2, 10, 14, 21, 22, 34, 35, 41].

The market-integration literature distinguishes long-run price convergence from shorter-run shock transmission. European evidence shows that integration varies across borders and market arrangements [17, 45]. Related work documents the transmission of shocks across electricity forward markets [4]. For the Iberian case, Ciarreta and Zarraga [6] examine mean and volatility transmission in MIBEL and compare it with the EPEX spot market. These studies establish that price co-movement is an important dimension of market integration. The present paper uses a narrower operational measure: the mean hourly absolute price difference between the Spanish and

Portuguese bidding zones. This measure records within-day market splitting that may be hidden by a comparison of daily mean prices.

Research on the Iberian Exception addresses a different form of market intervention. Linares and Gomez [26] describe the mechanism and assess its implications for electricity-price formation. Using a synthetic-control approach, Haro Ruiz et al. [19] examine its effects on wholesale prices and consumer inflation. More recent work considers how the energy crisis and the Iberian intervention redistributed costs and benefits between generators and consumer groups [13]. Together, these studies show that the observed market-clearing price did not represent all costs ultimately borne by consumers. The present analysis does not estimate the causal or distributional effect of the intervention. It instead tests whether the renewable–gas pass-through association differed during the formal policy window.

H4: *Greater internal Spain-Portugal interconnection capacity is associated with narrower day-ahead price spreads within MIBEL.*

H5: *The renewable–gas pass-through interaction varies across the pre-crisis, gas-crisis, Iberian Exception and post-crisis periods.*

4 Data and variables

The raw country-day dataset covers Spain and Portugal from 1 January 2019 to 31 December 2025. The 2019 start provides two complete pre-crisis years before the gas-price surge began in 2021. It also permits consistent daily matching of the price, generation, demand, gas and interconnection series used in all models. The period contains 2,557 calendar days and 5,114 potential country-day observations. Complete price records are available for 5,106 country-days. Two of these observations lack a matched MIBGAS day-ahead gas price. A further 346 lack both import and export commercial-capacity records. The matched baseline price and volatility sample therefore contains 4,758 country-day observations, or 2,379 dates observed for both countries. The internal price-spread model uses the same 2,379 dates. No partial 2026 observations enter the analysis. Given the two-country setting, the design should be read as a high-frequency market study rather than as a broad multi-country panel.

Table 1 summarises the data used in the empirical analysis. OMIE provides day-ahead prices and Spain-Portugal interconnection-capacity files [37]. REData provides Spain’s generation structure and demand [39]. REN Data Hub provides Portugal’s daily electricity balance [40]. MIBGAS annual public files provide day-ahead natural-gas prices for the Iberian gas market [29]. Figure 1 shows how these sources are converted into the estimation panel and robustness outputs (Table 2, Table 3).

Table 1 Data sources and variable blocks

The weather checks use country-day averages from eight representative grid points in Spain and four in Portugal. The remaining variables are constructed from the listed electricity-market, generation and gas-market sources

Data block	Source	Frequency	Use in paper
Electricity prices and internal interconnection	OMIE	Market-period/daily	Day-ahead price, within-day volatility, hourly Spain-Portugal spread, internal interconnection capacity
Spain generation and demand	Red Electrica / REData API	Daily	Renewable share, wind/solar/hydro shares, gas generation share, demand
Portugal generation and consumption	REN Data Hub	Daily	Renewable share, wind/solar/hydro shares, gas generation share, consumption
Natural gas prices	MIBGAS annual public files	Daily	PVB day-ahead gas price, gas shocks, Iberian Exception interaction checks
Weather conditions	NASA POWER Daily API	Daily	Temperature, precipitation, wind speed and solar radiation controls

Fig. 1 Data construction pipeline from raw market, generation and gas-price sources to the estimation panel and robustness outputs

Data and empirical workflow

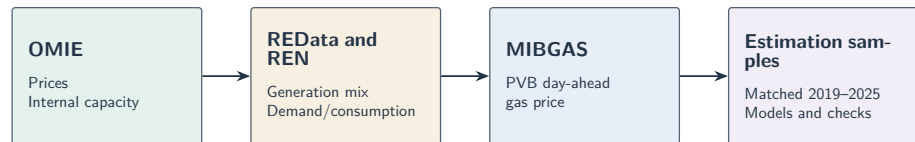


Table 2 Sample construction

Step	Observation unit	Observations
Calendar days, 2019–2025	days	2557
Potential Spain–Portugal country-days	country-days	5114
Country-day price and volatility observations	country-days	5106
Matched with generation and demand data	country-days	5106
Matched with MIBGAS day-ahead gas prices	country-days	5104
Final baseline price/volatility estimation sample	country-days	4758
Final internal price-spread estimation sample	days	2379

The raw files cover 2019–2025, but the estimation samples are matched samples after merging prices, generation, gas prices and interconnection-capacity variables. The final internal price-spread sample is a daily Spain–Portugal spread dataset, so its observation unit is days rather than country-days

Table 3 Comparison of included and excluded country-day observations

	Variable	Included baseline sample	Excluded: capacity missing	Excluded: gas price missing
The capacity-missing group contains observations for which both import and export commercial-capacity records are unavailable. The comparison is descriptive and does not establish that observations are missing at random	Day-ahead price (EUR/MWh)	81.660	95.220	47.967
	Renewable share (pp)	56.887	49.910	45.357
	Wind and solar share (pp)	34.497	30.375	34.401
	Hydro share (pp)	17.007	14.357	5.023
	Gas generation share (pp)	22.244	26.865	34.306
	MIBGAS day-ahead price (EUR/MWh)	40.561	39.357	
	Observations	4758	346	2
	Share in broad crisis period (%)	35.9	34.1	0.0

The dependent variables are the daily average day-ahead price, within-day price volatility and the internal Spain–Portugal price spread. The preferred spread variable is the mean of hourly absolute Spain–Portugal price differences, because it preserves within-day market splitting that can be hidden when only daily average prices are compared. The alternative daily-mean price spread is retained as a robustness check. The explanatory variables include renewable generation share, wind-and-solar generation share, gas generation share, log day-ahead gas price, log demand and import/export interconnection capacity. Import and export capacity enter the price-level specifications only as controls for internal transmission and market-coupling conditions. Their coefficients are not interpreted as structural price effects. The substantive integration mechanism is examined separately through the internal MIBEL price-spread model. Table 4 defines the variables and units, while Table 5 reports descriptive statistics.

The volatility checks also distinguish the level of gas prices from gas-price instability. Two supplementary controls are used. The first is the absolute daily change in the log gas price. The second is the rolling seven-day standard deviation of daily log changes. These measures are added separately to the volatility interaction model.

Daily weather measures come from the NASA POWER Daily API [33]. We use temperature, precipitation, wind speed and solar radiation for 2019–2025. Country-day values are equal-weighted averages across eight

Table 4 Variable construction and units

Variable	Definition	Unit	Aggregation
<i>Dependent variables</i>			
Day-ahead price	Daily average of all available OMIE day-ahead market periods for each country	EUR/MWh	Market-day mean
Within-day volatility	Standard deviation of all day-ahead market-period prices within a market day	EUR/MWh	Market-day standard deviation
Hourly absolute price spread	Mean of the hourly absolute Spain-Portugal day-ahead price difference	EUR/MWh	Hourly mean absolute difference
Daily mean price spread	Absolute difference between the Spanish and Portuguese daily average day-ahead prices	EUR/MWh	Absolute difference of daily means
<i>Main explanatory variables</i>			
Renewable share	Renewable domestic generation divided by total domestic generation	Percentage points	Daily share ×100
Wind and solar share	Wind plus solar PV for Spain; wind plus reported solar for Portugal, divided by total domestic generation	Percentage points	Daily share ×100
Hydro share	Reported hydro generation divided by total domestic generation; Portugal's Hydro 1 series excludes pumped-storage generation	Percentage points	Daily share ×100
Gas price	MIBGAS PVB day-ahead natural-gas price	EUR/MWh	Daily price; log in models
Gas-price instability	Absolute daily log change and seven-day rolling standard deviation of daily log changes	Log-price units	Daily robustness measures
Internal interconnection capacity	Daily average available Spain-Portugal commercial capacity, averaged across import and export directions	GW	Hourly capacity mean divided by 1,000
Iberian Exception	Dummy equal to one during the formal gas-price cap mechanism period, 15 June 2022 to 31 December 2023	0/1	Daily
<i>Controls</i>			
Gas generation share	Combined-cycle generation for Spain and reported natural-gas generation for Portugal, divided by total domestic generation	Percentage points	Daily share ×100
Temperature	Mean temperature at 2 m and its square	Degrees Celsius	Daily country mean
Precipitation	Corrected precipitation	mm/day	Daily country mean
Wind speed	Mean wind speed at 10 m	m/s	Daily country mean
Solar radiation	All-sky surface shortwave radiation	kWh m ⁻² day ⁻¹	Daily country mean
Demand and consumption	Electricity demand for Spain and reported consumption for Portugal	GWh	Daily total; log in models

Generation categories follow each country's official source classification. Renewable share includes hydro; the denominator for generation shares is domestic generation and excludes imports. The spread and capacity measures refer to internal Spain-Portugal integration within MIBEL, not external Iberia-France integration

representative grid points in Spain and four in Portugal. The weather file covers every country-day in the analysis panel. These variables are used in robustness checks, not as instruments.

Daily averages and standard deviations are computed using all available OMIE market periods for each market day. Most days have 24 hourly observations, while daylight-saving transitions and the 2025 shift to shorter market-time units generate days with 23, 25, 96 or 100 observations. These days are retained in the main construction, and the spread results are checked on 24-period days only. Table 6 reports the period-count distribution in the matched internal price-spread sample.

Table 5 Descriptive statistics

Country	Variable	N	Mean	Std. dev	Min	Max
Portugal	Day-ahead electricity price (EUR/MWh)	2553	82.791	60.235	0.105	542.779
Portugal	Within-day price volatility (EUR/MWh)	2553	18.032	13.744	0.252	113.187
Portugal	Renewable generation share (pp)	2553	64.901	16.925	18.103	96.774
Portugal	Wind and solar generation share (pp)	2553	34.978	15.732	2.793	76.667
Portugal	Gas generation share (pp)	2553	26.733	16.084	0.588	69.608
Portugal	Electricity demand/consumption (GWh)	2553	138.479	15.385	71.000	185.000
Portugal	MIBGAS PVB day-ahead gas price (EUR/MWh)	2552	40.479	33.512	4.140	241.360
Portugal	Net export to other Iberian market (MW)	2380	-813.647	1082.063	-3391.550	3053.058
Portugal	Import capacity from other Iberian market (MW)	2380	3421.990	725.668	0.000	4943.750
Portugal	Export capacity to other Iberian market (MW)	2380	3087.876	413.231	0.000	4419.375
Spain	Day-ahead electricity price (EUR/MWh)	2553	82.342	60.334	0.440	544.984
Spain	Within-day price volatility (EUR/MWh)	2553	18.596	14.130	0.542	113.187
Spain	Renewable generation share (pp)	2553	47.919	11.105	19.656	75.127
Spain	Wind and solar generation share (pp)	2553	33.458	11.129	7.054	63.044
Spain	Gas generation share (pp)	2553	18.390	8.629	3.824	48.957
Spain	Electricity demand/consumption (GWh)	2553	693.636	70.869	403.297	875.119
Spain	MIBGAS PVB day-ahead gas price (EUR/MWh)	2552	40.479	33.512	4.140	241.360
Spain	Net export to other Iberian market (MW)	2380	813.647	1082.063	-3053.058	3391.550
Spain	Import capacity from other Iberian market (MW)	2380	3087.876	413.231	0.000	4419.375
Spain	Export capacity to other Iberian market (MW)	2380	3421.990	725.668	0.000	4943.750

Generation shares are reported in percentage points to match the regression tables and coefficient interpretation

Table 6 Market-period counts in the matched internal price-spread sample

Market periods per day	Number of days	Interpretation
23	7	DST or 15-minute market-time-unit day
24	2278	Standard hourly day
25	4	DST or 15-minute market-time-unit day
96	89	DST or 15-minute market-time-unit day
100	1	DST or 15-minute market-time-unit day

The count distribution refers to the final matched internal price-spread and interconnection-capacity sample used in the spread regression, not the full price-only country-day file. Daily averages and standard deviations use all available market periods. Robustness checks include the 24-period-day-only spread specification

5 Empirical strategy

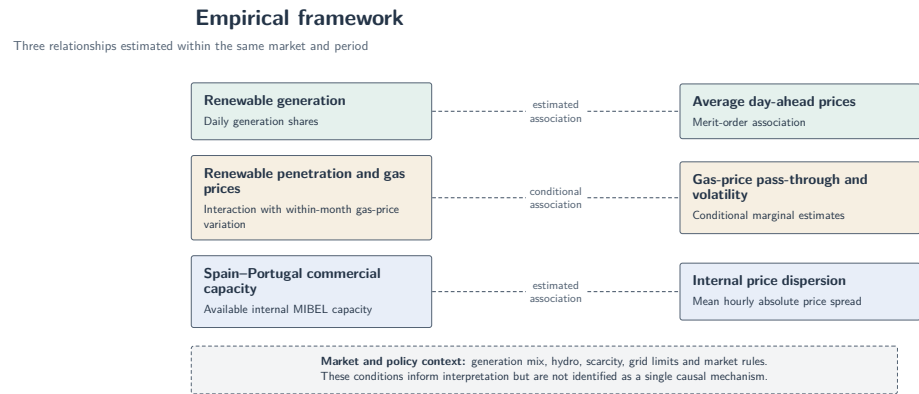
We estimate a sequence of linear fixed-effects regressions by ordinary least squares. Models M1–M4 use country-day observations, while Model M5 uses one Spain–Portugal price-spread observation per day.

The empirical design is intentionally associative. Daily renewable generation is partly weather-driven. It can move with wind speed, solar irradiation, temperature, hydro inflows, reservoir conditions and demand. These factors may also affect electricity prices. Demand, generation shares and interconnection conditions may be jointly determined with prices through scarcity, outages and market expectations. Country, year-month and weekday fixed effects absorb persistent country differences, common monthly shocks and recurring weekly patterns. They do not isolate exogenous renewable-availability shocks. The robustness checks add observed daily weather conditions and examine alternative inference, lagged regressors, VRE-only specifications and technology decomposition. They do not turn the estimates into structural causal effects.

Figure 2 summarises the empirical relationships. It does not represent a linear causal chain.

Several regressors require particular caution. Demand is the quantity recorded for the market day and is part of the market-clearing environment. Renewable and gas shares are realised generation outcomes. They can respond to weather, availability and expected prices. Import and export capacity measure available commercial capacity

Fig. 2 Empirical relationships examined in the paper. Dashed lines denote estimated associations, not identified causal effects



rather than realised flows. Even so, outages and network conditions may affect both capacity and prices. These variables should therefore be treated as contemporaneous market controls, not as exogenous treatments.

The data do not provide a credible external instrument for renewable output, demand or capacity. Daily weather controls are available, but plant outages, hydro inflows and reservoir conditions are not consistently observed for both countries. We therefore do not estimate an instrumental-variable model. Weather can affect prices through demand and scarcity as well as through renewable output, so an exclusion restriction would not be credible here. A separate check lags the regressors by one day. This reduces direct contemporaneous overlap, but it does not remove omitted-variable bias or establish causality.

Let $g_t = \log(\text{GasPrice}_t)$, and let $\bar{g}$ denote the sample mean of g_t . The centred gas-price variable is defined as:

$$\tilde{g}_t = g_t - \bar{g}.$$

The control vector is

$$Z'_{ct}\gamma = \gamma_1 \log(\text{Demand}_{ct}) + \gamma_2 \text{GasShare}_{ct} + \gamma_3 \text{ImportCapacity}_{ct} + \gamma_4 \text{ExportCapacity}_{ct}.$$

Thus, import and export capacity enter separately. In the equations below, α is the intercept. The symbols β , ϕ and θ denote slope coefficients, while γ contains the control coefficients. The terms ϵ_{ct} , ν_{ct} and u_t are regression errors. The term μ_c denotes country fixed effects. The terms $\lambda_{ym(t)}$ and $\delta_{d(t)}$ denote year-month and weekday fixed effects.

Model M1 is the baseline price regression:

$$\text{Price}_{ct} = \alpha + \beta_1 \text{Renewable}_{ct} + \beta_2 \tilde{g}_t + Z'_{ct}\gamma + \mu_c + \lambda_{ym(t)} + \delta_{d(t)} + \epsilon_{ct}. \quad (1)$$

Here, c denotes country and t denotes day. Model M2 uses the same controls but replaces the dependent variable with within-day price volatility:

$$\text{Volatility}_{ct} = \alpha + \beta_1 \text{Renewable}_{ct} + \beta_2 \tilde{g}_t + Z'_{ct}\gamma + \mu_c + \lambda_{ym(t)} + \delta_{d(t)} + \nu_{ct}. \quad (2)$$

Mean-centring does not change model fit or an interaction slope. In interaction models, it makes the renewable coefficient interpretable at the sample-mean log gas price. The capacity variables control for contemporaneous internal congestion conditions. Their coefficients are not treated as the H4 integration estimate. Demand is included as a market-condition control. Its coefficient is not interpreted causally. Country fixed effects absorb time-invariant differences between Spain and Portugal. Year-month fixed effects absorb common monthly shocks. Weekday fixed effects control recurring demand and market patterns. Country-by-date fixed effects are not used

because they would absorb all country-day regressors. The gas-price coefficient is identified from daily variation within each month. Alternative year and year-week fixed effects are examined in robustness checks.

Generation shares are compositional because their total is constrained. An increase in one share must coincide with a decrease in one or more other categories. A coefficient should therefore be read relative to the omitted generation mix. It does not describe an isolated increase that leaves every other share unchanged. A model without gas share estimates the overall renewable-price association. A model with gas share estimates a conditional association at a fixed gas share. A technology-decomposition model separates wind, solar, hydro and gas shares. Other generation and net balancing sources form the omitted category.

Model M3 tests gas-price pass-through:

$$\begin{aligned} Price_{ct} = & \alpha + \beta_1 Renewable_{ct} + \beta_2 \tilde{g}_t + \beta_3 Renewable_{ct} \\ & \times \tilde{g}_t + Z_{ct}'\gamma + \mu_c + \lambda_{ym(t)} + \delta_{d(t)} + \epsilon_{ct}. \end{aligned} \quad (3)$$

The coefficient β_3 is the main pass-through parameter. A negative value means that the marginal gas-price estimate is smaller when renewable penetration is higher.

$$\begin{aligned} Volatility_{ct} = & \alpha + \beta_1 VRE_{ct} + \beta_2 \tilde{g}_t + \beta_3 VRE_{ct} \times \tilde{g}_t \\ & + Z_{ct}'\gamma + \mu_c + \lambda_{ym(t)} + \delta_{d(t)} + \nu_{ct}. \end{aligned} \quad (4)$$

This is Model M4. VRE_{ct} is the combined wind-and-solar generation share. Its interaction coefficient shows how the marginal VRE estimate varies with the gas-price level.

Because the Iberian Exception changed crisis-period price formation, an additional specification interacts the gas-price and renewable pass-through terms with the formal mechanism period from 15 June 2022 to 31 December 2023. The standalone Iberian Exception dummy is absorbed by the time fixed effects. The specification therefore focuses on changes in gas-price pass-through:

$$\begin{aligned} Price_{ct} = & \alpha + \beta_1 Renewable_{ct} + \beta_2 \tilde{g}_t + \beta_3 Renewable_{ct} \\ & \times \tilde{g}_t + \phi_1 \tilde{g}_t \times Exception_t + \phi_2 Renewable_{ct} \times \tilde{g}_t \\ & \times Exception_t + Z_{ct}'\gamma + \mu_c + \lambda_{ym(t)} + \delta_{d(t)} + \epsilon_{ct}. \end{aligned} \quad (5)$$

Model M5 examines internal MIBEL integration:

$$\begin{aligned} HourlyAbsSpread_t = & \alpha + \theta_1 Interconnection_t + \theta_2 AvgRenewable_t \\ & + \theta_3 RenewableGap_t + \theta_4 AvgGasShare_t \\ & + \theta_5 \tilde{g}_t + \lambda_{ym(t)} + \delta_{d(t)} + u_t. \end{aligned} \quad (6)$$

Here, $Interconnection_t$ is the daily average available Spain-Portugal commercial capacity. It is averaged across directions and expressed in GW. $AvgRenewable_t$ and $AvgGasShare_t$ are the two-country daily averages. $RenewableGap_t$ is the absolute difference between the Spanish and Portuguese renewable shares. Unlike the capacity controls in M1–M4, $Interconnection_t$ is paired with the hourly absolute price spread to test internal integration directly.

The main tables use HC1 standard errors [43]. HC1 is a heteroskedasticity-consistent covariance estimator with a finite-sample degrees-of-freedom adjustment. It allows the residual variance to differ across observations. The first inference check clusters by market date [1, 5]. This permits the Spanish and Portuguese residuals to be correlated within the same day. It addresses common daily shocks, but not serial dependence across dates.

The second check uses a time-series heteroskedasticity-and-autocorrelation-consistent estimator [3, 36]. Daily score contributions are first aggregated across countries. The estimator then allows these daily contributions to be heteroskedastic and serially correlated. A seven-day lag covers one weekly market cycle. A 30-day lag checks whether inference remains stable under longer persistence. These are transparent sensitivity windows rather than data-selected optimal bandwidths. Country-clustered inference is not used because the panel contains only two country clusters.

Table 7 Main regression results

Variable	M1 price	M2 volatility	M3 price interaction	M4 volatility interaction
Renewable share (pp)	−0.699*** (0.064)	−0.026 (0.026)	−0.861*** (0.072)	
Wind and solar share (pp)				0.010 (0.016)
Mean-centred log gas price, $\bar{g}_t$	68.993*** (4.794)	10.231*** (1.302)	91.070*** (5.558)	6.919*** (1.459)
Renewable share x $\bar{g}_t$			−0.438*** (0.047)	
Wind/solar share x $\bar{g}_t$				0.101*** (0.017)
Log demand	53.853*** (5.407)	−12.360*** (2.204)	62.275*** (5.357)	−11.956*** (2.243)
Gas generation share (pp)	0.351*** (0.068)	−0.212*** (0.027)	0.119 (0.076)	−0.168*** (0.019)
Import capacity (GW)	3.152*** (0.669)	1.073*** (0.291)	4.300*** (0.676)	1.099*** (0.287)
Export capacity (GW)	5.118*** (0.638)	0.883*** (0.287)	3.998*** (0.650)	0.944*** (0.284)
Observations	4758	4758	4758	4758
Adjusted R2	0.888	0.663	0.892	0.667

Robust standard errors are reported in parentheses where applicable. The gas-price variable is $\bar{g}_t = g_t - \bar{g}$, where $g_t = \log(\text{GasPrice}_t)$ and $\bar{g}$ is the sample mean of g_t . Import and export capacity are country-day controls; the preferred internal MIBEL price-spread model is reported separately in Table 12. *, **, and *** denote significance at the 10%, 5%, and 1% levels

Table 8 Generation-share composition checks

Variable	A: Total renewable association	B: Conditional on gas share	C: Technology decomposition
Renewable share (pp)	−0.987*** (0.035)	−0.699*** (0.064)	
Wind share (pp)			−0.684*** (0.062)
Solar share (pp)			−0.282** (0.135)
Hydro share (pp)			−0.002 (0.073)
Gas generation share (pp)		0.351*** (0.068)	0.409*** (0.065)
Mean-centred log gas price, $\bar{g}_t$	69.164*** (4.799)	68.993*** (4.794)	68.626*** (4.799)
N	4758	4758	4758
Adjusted R2	0.887	0.888	0.896

All models include country, year-month and weekday fixed effects. Model A omits gas share and estimates the total renewable-price association; Model B conditions on gas share; Model C decomposes renewables into wind, solar and hydro. The omitted generation category in Model C is other generation and net balancing sources

6 Results and discussion

The results cover price levels, gas-price pass-through, volatility, internal price dispersion, period heterogeneity and robustness.

6.1 Renewables, electricity prices and gas pass-through

Table 7 reports the price and volatility specifications. In the price equation that conditions on gas generation, the renewable-share coefficient is −0.699 (standard error 0.064, $N = 4758$). Interpreted at the daily frequency, a one percentage point higher renewable share corresponds to a 0.699 EUR/MWh lower day-ahead price, holding gas share and other controls fixed. The total-association specification in Table 8, which does not condition on gas share, is larger at −0.987 EUR/MWh. The technology decomposition shows that wind and solar carry most of the negative association, while the hydro coefficient is close to zero. Gelabert et al. [15] also report a negative renewable-price association for Spain. Sensfuss et al. [42] and Ketterer [23] document related merit-order patterns in Germany. The present estimates differ because they combine Spain and Portugal and condition on gas generation and internal commercial capacity. They support H1 as an association, not as a causal estimate of renewable entry.

The gas-price coefficient remains large even in a high-renewable system. With year-month fixed effects, the coefficient on $\bar{g}_t$ is identified from daily gas-price movements within the same month; in the baseline price model

Table 9 Technology-specific gas-price pass-through

Variable	Coefficient (HC1 SE)
Wind and solar share (pp)	−0.823*** (0.067)
Hydro share (pp)	−0.102 (0.078)
Mean-centred log gas price	89.090*** (5.516)
Wind and solar share × gas price	−0.614*** (0.051)
Hydro share × gas price	−0.007 (0.063)
Observations	4758
Adjusted R^2	0.904

HC1 standard errors in parentheses. Wind-and-solar and hydro shares enter together with their interactions with the mean-centred log gas price. Controls include log demand, gas-generation share, import and export capacity, country fixed effects, year-month fixed effects and weekday fixed effects. The specification does not identify exogenous hydro scarcity. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 10 Marginal VRE estimate for within-day price volatility

Gas-price point	Gas price (EUR/MWh)	Log gas price	Marginal VRE estimate
25th percentile	17.200	2.845	−0.048*** (0.013)
Median	32.305	3.475	0.016 (0.017)
75th percentile	46.207	3.833	0.052** (0.021)
95th percentile	115.291	4.747	0.145*** (0.035)
Sample maximum	241.360	5.486	0.220*** (0.046)
Zero-estimate threshold	27.593	3.318	0.000

The marginal estimate is computed as $\beta_{VRE} + \beta_{int}(g_t - \bar{g})$, where $g_t = \log(GasPrice_t)$. HC1 delta-method standard errors are in parentheses

it is 68.993 (standard error 4.794), supporting H2a. This positive association relates to the fuel-price transmission documented by Mohammadi [30] and Mohammadi [31]. It also accords with the evidence in Zakeri et al. [46] that gas-fired units can set European electricity prices more often than their generation share suggests. The present model adds a renewable interaction. The renewable coefficient is −0.861 at the sample-mean gas price, and $Renewable_{ct} \times \bar{g}_t$ is −0.438 (standard error 0.047). The negative sign indicates that days with higher renewable penetration also display a weaker price response to within-month gas-price variation. Identifying changes in the marginal generating technology would require unit-level bidding data.

The technology-specific pass-through model separates wind and solar from hydro generation. The wind-and-solar interaction with gas prices is −0.614 (standard error 0.051). The hydro interaction is −0.007 (standard error 0.063) and is not statistically significant. The full-sample moderation estimate is therefore concentrated in wind and solar generation. This result does not identify exogenous hydro scarcity because hydro generation remains an observed market outcome (Table 9).

6.2 Price volatility

The volatility results require a cautious interpretation. Gas prices have a positive coefficient in the volatility model, in line with H2b. For wind and solar, the baseline marginal estimate varies with the level of gas prices. The interaction coefficient is 0.101 (standard error 0.017). Table 10 and Fig. 3 show a negative baseline estimate at the 25th percentile, an estimate close to zero around the median, and positive estimates from the 75th percentile upward. The implied baseline zero threshold is about 27.6 EUR/MWh. Woo et al. [44] link wind generation to both the level and variance of spot prices, while Macedo et al. [28] examine volatility in the Iberian setting. The present result is narrower. It shows that the estimated VRE–volatility association changes with the gas-price level.

Table 11 presents two checks for gas-price instability. The absolute daily log change has a positive coefficient of 5.284 and is significant at the 10% level. The seven-day volatility measure has a coefficient of 16.248 and is significant at the 1% level. The wind-and-solar interaction remains close to the baseline estimate in both checks. It is 0.102 with the absolute-change control and 0.104 with the seven-day measure. Research on flexibility,

Fig. 3 Marginal wind-and-solar estimate for within-day price volatility at selected gas-price levels

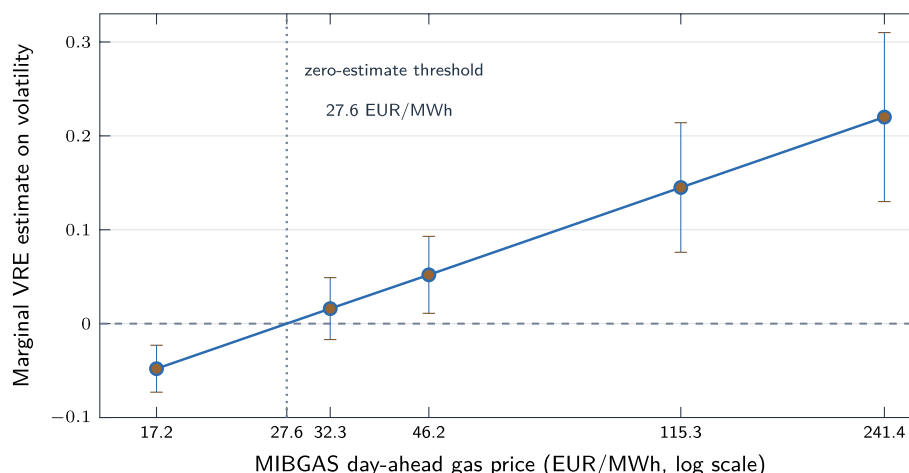


Table 11 Gas-price instability checks for within-day electricity-price volatility

Specification	VRE × gas-price level	Gas-instability control	N
Baseline volatility interaction	0.101*** (0.017)		4758
Absolute daily log gas-price change	0.102*** (0.017)	5.284* (2.803)	4754
Seven-day volatility of log gas-price changes	0.104*** (0.017)	16.248*** (5.040)	4748

The dependent variable is within-day electricity-price volatility. All specifications include the gas-price level, wind-and-solar share, log demand, gas-generation share, import and export capacity, country fixed effects, year-month fixed effects and weekday fixed effects. HCl standard errors are in parentheses. The seven-day measure is the rolling standard deviation of daily log gas-price changes. *, **, and *** denote significance at the 10%, 5%, and 1% levels

Table 12 Internal MIBEL price-spread robustness

Specification	Internal interconnection capacity (GW)	Average renewable share (pp)	Renewable share gap (pp)	Mean-centred log gas price, $\bar{g}_t$	N	Adjusted R ²
Mean hourly absolute spread	-2.924*** (0.604)	-0.031* (0.017)	0.004 (0.008)	0.569 (0.520)	2379	0.244
Abs. spread of daily mean prices	-2.468*** (0.413)	-0.034** (0.017)	0.004 (0.008)	0.547 (0.519)	2379	0.230
Mean hourly absolute spread, 24-period days only	-2.973*** (0.620)	-0.030* (0.018)	0.003 (0.008)	0.588 (0.523)	2278	0.246

The preferred dependent variable is the mean hourly absolute Spain-Portugal price spread. The daily-mean spread is shown as a robustness check

storage and balancing explains why these conditions can matter in renewable systems [11, 16, 25, 27]. The present specification estimates the VRE–gas-price interaction; it leaves these system mechanisms open (Table 12).

Supplementary Tables S2–S3 report the weather-control checks. The renewable–gas interaction remains close to the baseline estimate. The price-level coefficient becomes smaller. The baseline low-gas volatility result is not robust to the full weather specification.

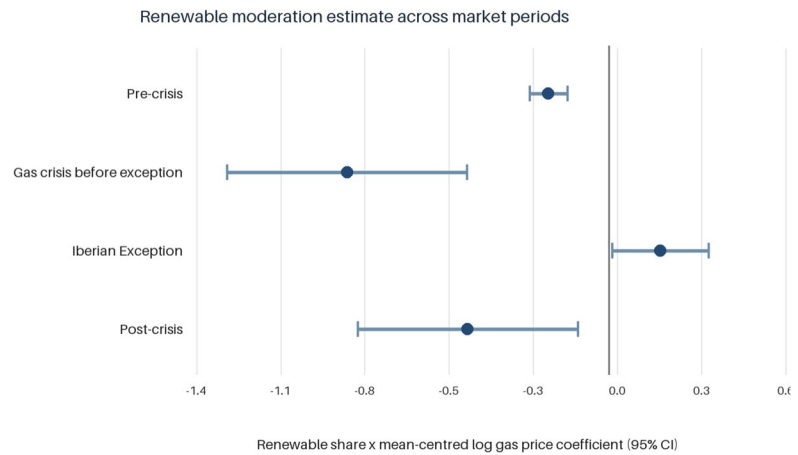
6.3 Internal MIBEL price integration

The spread model provides the internal-coupling result. In the preferred hourly-spread specification, the coefficient on average Spain-Portugal commercial capacity is -2.924 (standard error 0.604, $N = 2379$). One GW higher available internal capacity is therefore associated with a 2.924 EUR/MWh smaller mean hourly absolute price difference between the two bidding zones. This estimate is distinct from the import and export capacity controls in the country-day price equations. It remains negative when the dependent variable is the absolute difference between daily mean prices and when the sample is restricted to 24-period days. The result supports H4, but only for internal MIBEL coupling.

Table 13 Renewable-gas pass-through across four market periods

Period	Renewable share (pp)	Mean-centred log gas price	Renewable share x gas price	N	Adjusted R2
Pre-crisis	−0.461*** (0.045)	33.718*** (2.564)	−0.206*** (0.033)	1592	0.879
Gas crisis before exception	0.205 (0.275)	275.555*** (14.584)	−0.889*** (0.208)	580	0.932
Iberian Exception	−1.664*** (0.209)	43.134*** (5.188)	0.173** (0.084)	1128	0.790
Post-crisis	−0.788*** (0.152)	108.113*** (14.862)	−0.480** (0.191)	1458	0.785

HC1 standard errors in parentheses. All specifications include log demand, gas-generation share, import and export capacity, country fixed effects, year-month fixed effects and weekday fixed effects. The periods describe temporal heterogeneity and do not identify the causal effect of the Iberian Exception. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Fig. 4 Renewable-gas interaction estimates across four market periods. Bars show 95% confidence intervals based on HC1 standard errors**Table 14** Country-specific and Iberian Exception robustness checks

Specification	Renewable share (pp)	$\bar{g}_t$	Renewable share x $\bar{g}_t$	N	Adjusted R2	$\bar{g}_t$ x Iberian Exception	Renewable x $\bar{g}_t$ x Iberian Exception
Full sample	−0.861*** (0.072)	91.070*** (5.558)	−0.438*** (0.047)	4758	0.892		
Spain only	−2.468*** (0.155)	101.102*** (9.109)	−0.969*** (0.095)	2379	0.915		
Portugal only	0.178* (0.099)	100.466*** (8.883)	−0.479*** (0.067)	2379	0.892		
With Iberian Exception interactions	−0.879*** (0.077)	130.052*** (8.038)	−0.592*** (0.060)	4758	0.902	−85.418*** (6.521)	0.774*** (0.102)

Country-specific models are estimated separately for Spain and Portugal. $\bar{g}_t = g_t - \bar{g}$, where $g_t = \log(\text{GasPrice}_t)$. The Iberian Exception specification interacts the mean-centred gas-price and renewable pass-through terms with the formal mechanism period from 15 June 2022 to 31 December 2023

6.4 Crisis-period heterogeneity

Table 13 separates four periods. The renewable-gas interaction is −0.206 before the crisis and −0.889 during the gas crisis before the Iberian Exception. It changes to 0.173 during the formal policy window and returns to −0.480 after 2023. Each estimate is statistically significant. Figure 4 shows the estimates and their 95% confidence intervals. The sign change is confined to the formal intervention period in this specification. It should not be read as a causal policy effect. The policy coincided with unusual scarcity, hydro and market conditions.

Table 14 shows a negative main interaction in both country samples. The larger Spain estimate may relate to differences in system size, solar generation and the role of combined-cycle plants. Portugal has a larger hydro share, but the models do not test whether this explains the country difference. The Iberian Exception specification also shows that the pass-through relationship changed during the formal intervention window. The policy,

however, coincided with unusual physical and market conditions. The estimates therefore do not identify its causal effect [19, 26].

6.5 Robustness checks

Table 15 reports the main sensitivity checks. The first four rows retain the baseline coefficient of -0.438 and change only the covariance estimator. Its standard error is 0.047 with HC1, 0.043 with date clustering, 0.061 with time-HAC lag 7 and 0.072 with time-HAC lag 30. Date clustering permits same-day correlation between the two countries. Time-HAC also permits serial dependence across days.

The lagged specification changes the timing of the regressors. Its interaction is -0.310 , compared with -0.438 in the contemporaneous model. This check is not an instrumental-variable correction. The 30-day gas-shock specification addresses short departures from the recent gas-price trend. The gas-shock coefficient is positive, but its interaction with renewable share is 0.003 and is not statistically significant. This alternative measure therefore does not reproduce the moderation found for the gas-price level. When wind-and-solar share replaces total renewable share, the interaction is -0.610 . The Spain and Portugal estimates are also negative, at -0.969 and -0.479 .

Table 16 changes the time controls. The interaction is -0.238 with year fixed effects, -0.351 with year-week fixed effects and -0.438 with the baseline year-month fixed effects. The magnitude varies because each model uses a different source of gas-price variation. The sign remains negative. Taken together, the checks show where the baseline result is stable and where it is sensitive. They do not remove the simultaneity concerns.

We also calculate variance inflation factors for the continuous regressors in Model M3. Each regressor is first residualised on the country, year-month and weekday fixed effects. The VIFs range from 1.06 to 4.54 . The maximum is 4.54 for gas-generation share. Table S4 reports the full diagnostic.

Table 15 Robustness checks

Robustness check	SE / sample	Key variable	Coefficient	N	Adjusted R ²
R1 main interaction	HC1	Renewable share x $\tilde{g}_t$	-0.438^{***} (0.047)	4758	0.892
R1 main interaction	Date-clustered	Renewable share x $\tilde{g}_t$	-0.438^{***} (0.043)	4758	0.892
R1 main interaction	time-HAC lag 7	Renewable share x $\tilde{g}_t$	-0.438^{***} (0.061)	4758	0.892
R1 main interaction	time-HAC lag 30	Renewable share x $\tilde{g}_t$	-0.438^{***} (0.072)	4758	0.892
R3 lagged regressors	HC1	Lagged renewable x $\tilde{g}_{t-1}$	-0.310^{***} (0.052)	4748	0.870
R3 lagged regressors	time-HAC lag 7	Lagged renewable x $\tilde{g}_{t-1}$	-0.310^{***} (0.057)	4748	0.870
R4 gas-shock interaction	HC1	30-day gas-price shock	0.513^{***} (0.153)	4758	0.890
R4 gas-shock interaction	HC1	Renewable share x gas-price shock	0.003 (0.002)	4758	0.890
R4 gas-shock interaction	time-HAC lag 7	30-day gas-price shock	0.513^{**} (0.258)	4758	0.890
R4 gas-shock interaction	time-HAC lag 7	Renewable share x gas-price shock	0.003 (0.003)	4758	0.890
R5 VRE specification	HC1	Wind/solar share x $\tilde{g}_t$	-0.610^{***} (0.044)	4758	0.904
R5 VRE specification	time-HAC lag 7	Wind/solar share x $\tilde{g}_t$	-0.610^{***} (0.066)	4758	0.904
R8 Spain only	time-HAC lag 7	Renewable share x $\tilde{g}_t$	-0.969^{***} (0.145)	2379	0.915
R8 Portugal only	time-HAC lag 7	Renewable share x $\tilde{g}_t$	-0.479^{***} (0.093)	2379	0.892

HC1 allows heteroskedasticity and applies a finite-sample adjustment. Date-clustered standard errors allow the Spanish and Portuguese residuals to be correlated within a market day. Time-HAC standard errors allow heteroskedasticity and serial correlation in daily score contributions. Lag 7 covers one weekly cycle; lag 30 checks longer persistence. $\tilde{g}_t = g_t - \bar{g}$, where $g_t = \log(\text{GasPrice}_t)$. The gas-shock check uses the percentage deviation of gas prices from their 30-day moving average. *, **, and *** denote significance at the 10%, 5%, and 1% levels

Table 16 Gas-price pass-through under alternative time fixed effects

Specification	Renewable share (pp)	Mean-centred log gas price, $\bar{g}_t$	Renewable share x $\bar{g}_t$	N	Adjusted R ²
Year FE	−0.903*** (0.096)	73.311*** (4.187)	−0.238*** (0.064)	4758	0.739
Year-week FE	−0.708*** (0.060)	110.743*** (6.500)	−0.351*** (0.039)	4758	0.935
Year-month FE	−0.861*** (0.072)	91.070*** (5.558)	−0.438*** (0.047)	4758	0.892

The interaction specifications use $\bar{g}_t = g_t - \bar{g}$, where $g_t = \log(\text{GasPrice}_t)$. The year-month specification identifies gas-price effects from within-month daily variation; the alternatives use broader year and year-week controls

6.6 Discussion

The price-level result extends the Iberian merit-order evidence of Gelabert et al. [15] to a later Spain–Portugal sample. The price-level estimate becomes smaller when wind and solar conditions are controlled for. The pass-through result addresses a different question. It shows that renewable penetration is associated with lower sensitivity to daily gas-price movements within a month. The interaction changes little after weather controls are added. This distinction matters because lower average prices do not necessarily imply lower fuel-price exposure. The data identify the price pattern, while the marginal generating technology remains unobserved.

The crisis-period results are the main qualification. The interaction remains negative during the gas crisis before the Iberian Exception. It changes sign during the formal policy period and becomes negative again after 2023. The technology model does not support a hydro-specific pass-through explanation in the full sample. These patterns narrow the set of plausible interpretations, but they do not isolate a policy effect. The Iberian Exception coincided with scarcity, changing hydro conditions and different compensation rules.

The internal-coupling result adds a spatial dimension. Zachmann [45] and Gugler et al. [17] study price convergence across European markets, while Ciarreta and Zarraga [6] examine mean and volatility transmission within MIBEL. The present outcome is more operational: the mean hourly absolute price difference between Spain and Portugal. Higher internal commercial capacity is associated with a smaller difference. This does not imply that stronger Iberia–France interconnection would have the same association, nor does it measure welfare gains from external integration.

The results have two practical implications for market monitoring. Average prices and gas-price sensitivity should be reported separately because they describe different dimensions of wholesale exposure. During interventions such as the Iberian Exception, the market-clearing price should also be read together with compensation payments. Internal commercial capacity provides a second indicator. Publishing its availability and outages alongside price spreads would make episodes of market splitting easier to interpret.

Comparisons of storage, demand response, flexible generation and balancing arrangements lie outside the empirical design. Earlier studies examine these mechanisms [16, 25, 27]. The Iberian Exception also separated part of the gas compensation from the market-clearing price. Consumer-cost analysis would need to add those compensation payments.

The wholesale–retail distinction further limits the social interpretation of the results. Contracts, regulated tariffs, taxes, hedging and public compensation can delay or redistribute wholesale-price changes. Wholesale moderation may ease one source of cost pressure without producing equal benefits for all households and firms. The analysis contains no retail-tariff, household-expenditure or income data. It therefore does not estimate affordability, energy-poverty or energy-justice outcomes.

6.7 Limitations

The evidence has several limits. First, the models identify conditional associations in a high-frequency two-country panel. They do not estimate a structural welfare effect. Second, the weather measures are averages of representative grid points. They are not weighted by population, renewable capacity or hydrological catchment. Hydro inflows and reservoir conditions are also unavailable on a consistent two-country basis. Third, simultaneity

remains possible. Demand and generation shares are part of the same market-clearing process as prices. Available commercial capacity can also reflect outages or network conditions that affect prices. Daily weather controls reduce one source of omitted variation, but they do not provide an exclusion restriction or remove joint determination. No credible external instrument is available in the dataset. The corresponding coefficients should not be read as causal responses. Fourth, the generation-share variables are compositional, so their coefficients depend on the included and omitted categories. Fifth, 346 country-day observations are excluded because commercial-capacity records are missing. These observations have higher average prices and lower renewable shares than the estimation sample. The missingness may therefore affect sample coverage. Sixth, the 2019–2025 window does not cover earlier energy-price episodes. A longer series could test whether the estimates extend to those periods. Such an extension would also need to reconcile historical source classifications and changes in market rules. Seventh, weekday fixed effects do not absorb every country-specific national holiday. These dates may create short-lived demand differences. Eighth, the outcomes are wholesale prices. Retail tariffs, contracts and public compensation may alter their transmission to consumers and firms. The study does not estimate household affordability, energy poverty, downstream inflation or industrial output. These limits are why the paper refers to wholesale price resilience rather than aggregate economic resilience.

7 Conclusion and policy implications

This paper used matched daily data for Spain and Portugal from 2019 to 2025. Fixed-effects regressions examined day-ahead price levels, within-month gas-price pass-through and internal MIBEL price dispersion. Renewable generation is associated with lower day-ahead prices. Renewable penetration is also associated with weaker gas-price pass-through. The latter estimate remains stable after daily weather controls are added. The interaction changes sign during the formal Iberian Exception period and becomes negative again after 2023. Internal commercial capacity is associated with narrower hourly price spreads.

The contribution is to examine these three price dimensions within the same paired market and period. Average prices alone do not show whether exposure to daily gas-price movements is weaker. Daily mean prices can also hide hourly separation between the Spanish and Portuguese bidding zones. Combining the outcomes reveals these differences without treating them as a single causal mechanism.

The results support concrete reporting priorities. Gas-price sensitivity should be monitored alongside average wholesale prices. During market interventions, compensation payments should accompany the reported clearing price. Internal capacity availability and outages should also accompany measures of Spain–Portugal price dispersion. Future work can extend the analysis with unit-level bids, hydro inflows, reservoir levels and direct measures of storage and balancing. Retail-tariff and compensation data are needed to test how the wholesale patterns reach households and firms.

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Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable. This study did not involve human participants, animals or biological materials. Not applicable.

Competing interests The authors declare no competing financial or non-financial interests.

Consent for publication Not applicable.

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